MERN STACK

DAY – 22 20 July 2026

MongoDB

POST Method

The POST method is an HTTP request method used to send data from client to server. It is mainly used for creating new records in a database.

Syntax:

```
app.post(path, callback_function)
```

Parameters:

Parameter	Description
path	URL route where request is sent
callback_function	Function executed when the route receives a POST request

Request Object (req)

The request object contains information sent by the client to the server.

Syntax:

```
req.body.property_name
```

Response Object (res)

The response object is used by the server to send data back to the client.

```
res.send({
 statuscode:1,
 msg:"Success"
})
```

The server sends a response message.

Syntax:

```
res.send(data)
```

Async and Await

async and await are JavaScript features used to handle asynchronous operations.

Database operations take time, so we use await to wait until MongoDB completes the operation.

Syntax:

```
async function functionName(){
   await asynchronous_operation();
}
```

Mongoose Model

A Mongoose model is an object that provides an interface to communicate with MongoDB collections.

Syntax:

```
mongoose.model(collectionName, schema)
```

findOne() Function

findOne() is a Mongoose method used to search a single document in MongoDB.

Syntax:

```
Model.findOne(condition)
```

save() Function

The save() method inserts a new document into MongoDB.

Syntax:

```
document.save()
```

Status Code Response

The API sends custom status codes:

```
Success:
{
```

```
statuscode:1,
msg:"Success"
}
```

Data inserted successfully

```
Duplicate Email:
{
statuscode:2,
msg:"Mail already exist"
}
```

Email is already present in database

```
Error:
{
statuscode:0,
msg:"error"
}
```

Insertion failed

```
export const Pd=()=>{
    const [name,setname]=useState("");
    const [age,setage]=useState("");
    const [mail,setmail]=useState("");
    const sub=async ()=>{
        const data=await fetch("http://localhost:9000/api/postuser",{
            method:"post",
            body:JSON.stringify({name,mail,age}),
            headers:{
                "Content-Type":"application/json"
            }
        })
        if (data.ok){
            let res=await data.json();
            if(res.statuscode === 1){
                console.log(res.msg);
            }else if(res.statuscode===2){
                alert(res.msg);
            }
            else{
                console.log(res.msg)
            }
        }
    }
    return(
        <>
            <input type="text" placeholder="enter name" onChange={(e)=>setname(e.target.value)}></input>
            <input type="number" placeholder="enter age"  onChange={(e)=>setage(e.target.value)}></input>
            <input type="email" placeholder="enter email"  onChange={(e)=>setmail(e.target.value)}></input>
            <button onClick={sub}>Click</button>
        </>
    );
}
```

```
const usermodel=mongoose.model("Detail",schema,"Detail")

app.post("/api/postuser",async(req,res)=>{
const findser=await usermodel.findOne({Email:req.body.mail});
if(findser){
    res.send({statuscode:2,msg:"Mail aleready exist"})
}else{

    const data=await usermodel({
        Name:req.body.name,
        Email:req.body.mail,
        Age:req.body.age
    });
    const savedata=await data.save();
    if(savedata){
        res.send({statuscode:1,msg:"Success"})
    }else{
        res.send({statuscode:0,msg:"error"})
    }
}
})
```

```
_id: ObjectId('6a5d9ff66de9db4349259589')
Name : "Aditi Tangri"
Email : "adititangri5@gmail.com"
Age : 20
__v : 0
```